BLENDER ADD-ON FOR THE LIVE FOR SPEED VEHICLE EDITOR

LFS SRE Tools

Open vehicle-editor `.sre` cars in Blender, edit them with real modelling tools, see them in the editor's own light, and export them back.

VERSION	BLENDER	LICENCE
2.0.6	4.2 or newer, made on 5.2	GPL-3.0-or-later

AUTHOR

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Fig. 1 An imported car in Blender's Material Preview. The LFS look lights it like the vehicle editor: same sun, sky reflections, paint and floor.

IN ONE MINUTE

Quick start

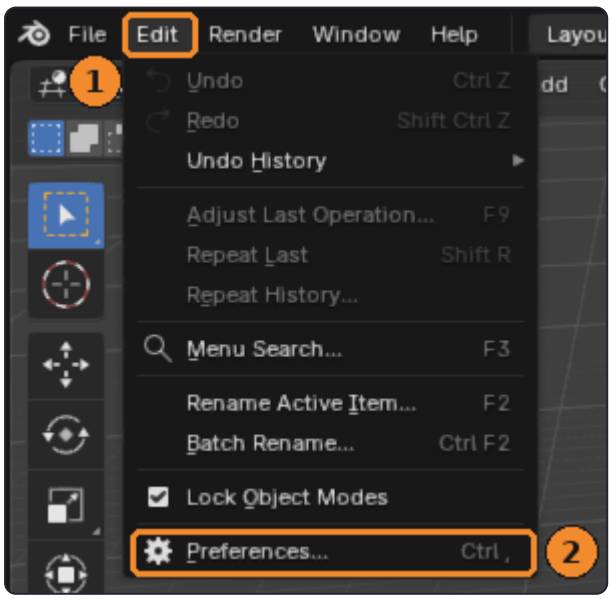
1. Install it with updates (Blender 4.2 or newer): **Preferences › Get Extensions**, the **Repositories** button at the top right, **+**, **Add Remote Repository**, paste `https://byemk.github.io/lfs-sre-tools/index.json`, **Create**, then search *LFS* and **Install**. Install has a picture of every click.
2. **File › Import › LFS SRE (.sre)** and pick a car from your LFS `data\3dob` folder.
3. Textures missing? Pick your LFS `data\png` folder in the box the LFS tab shows, and import again.
4. Edit in Solid view. Material Preview shows the car in the LFS editor's light.
5. **File › Export › LFS SRE** into `data\3dob` and open it in the vehicle editor. A file you write over is kept as `.sre.bak`.
6. Something wrong? **Copy Bug Report** at the bottom of the LFS SRE panel and paste it where you ask for help.

START HERE

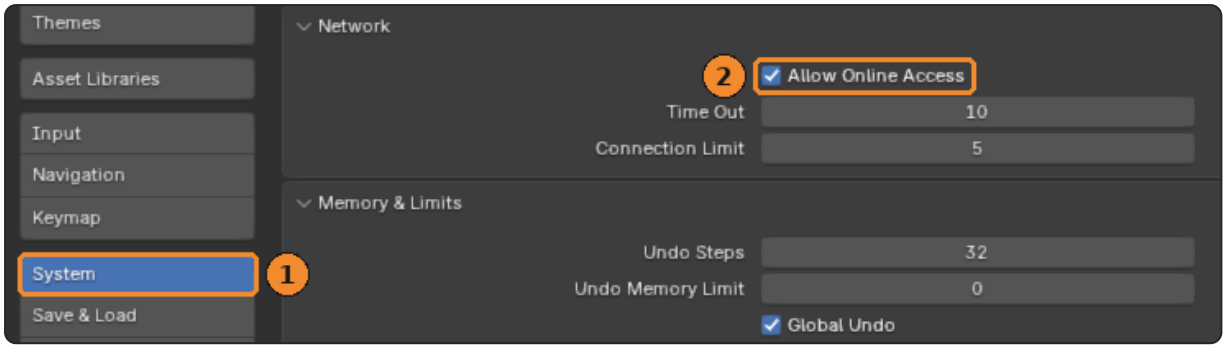
Install

You need Blender 4.2 or newer; it is tested on 4.2 LTS and 5.2. Install it from its update link and Blender keeps it up to date by itself. The orange numbers in the pictures are the clicks, in order.

1. Click **Edit** 1, then **Preferences** 2.



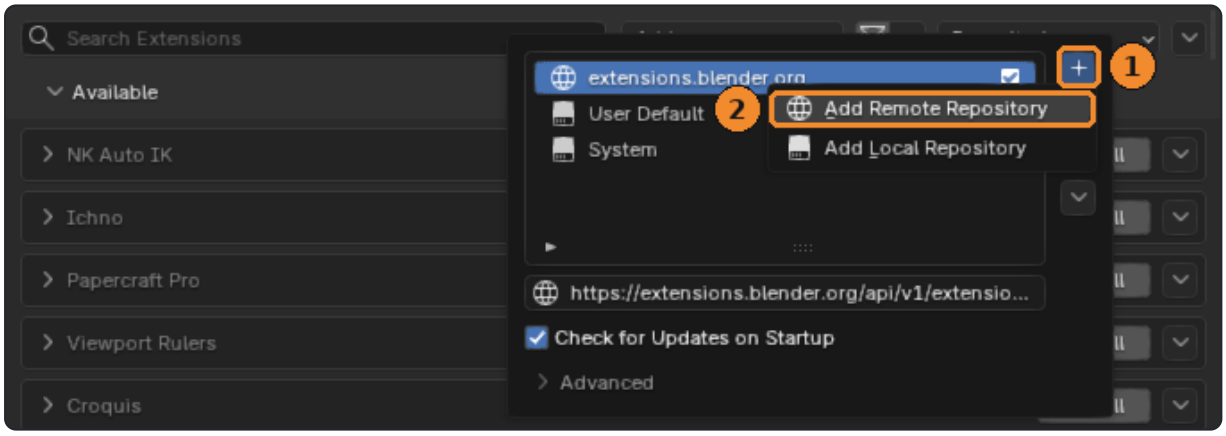
2 Click **System** 1 and tick **Allow Online Access** 2 under Network. Blender needs it to download extensions and to look for updates.



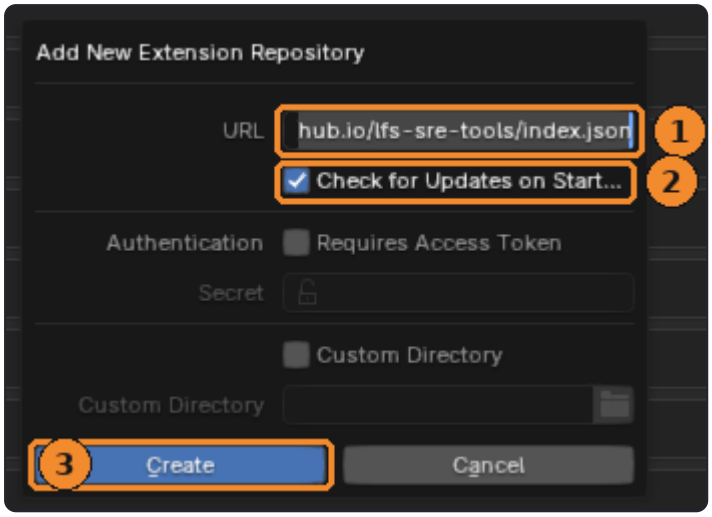
3 Click **Get Extensions** 1, then the **Repositories** button at the top right 2.



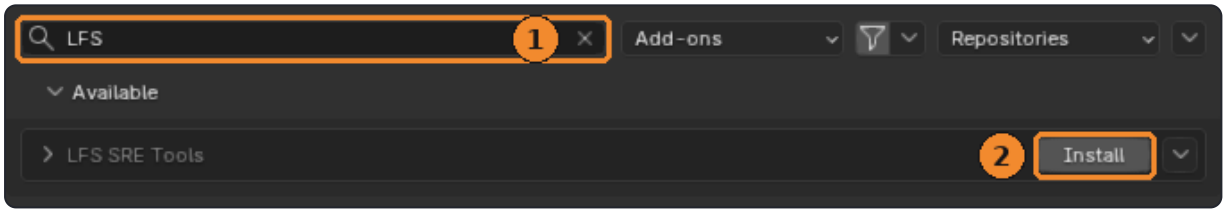
4 Click **+** 1 and choose **Add Remote Repository** 2.



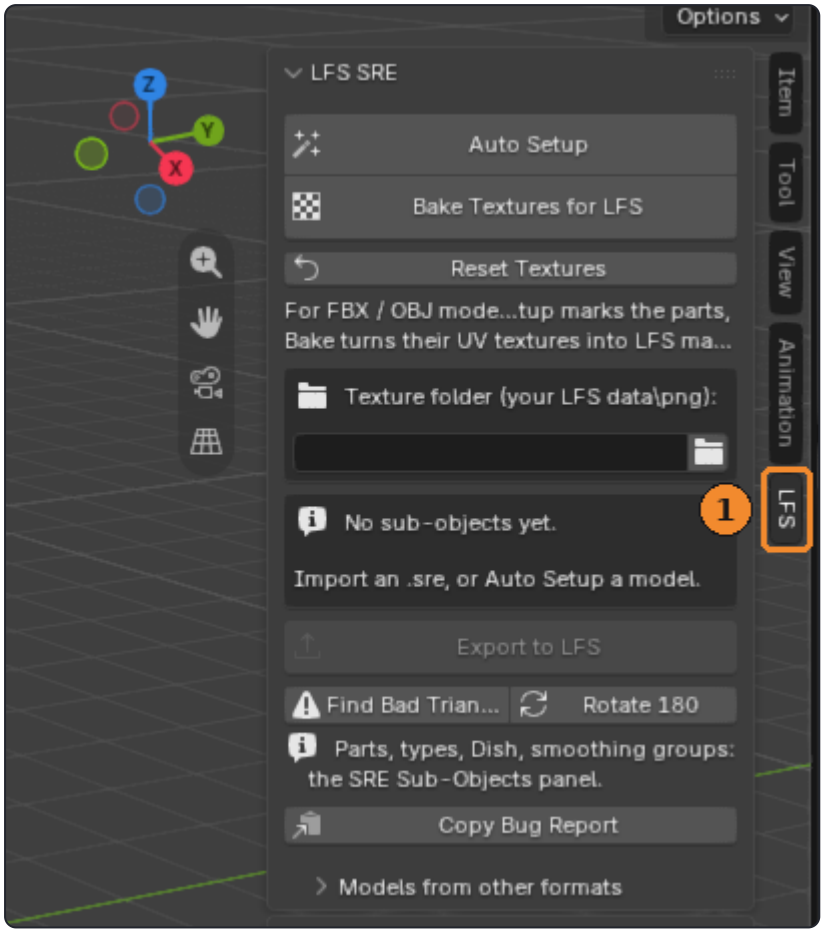
5 Paste `https://byemk.github.io/lfs-sre-tools/index.json` into **URL** (1), tick **Check for Updates on Startup** (2) and click **Create** (3).



6 Type *LFS* in the search box (1) and click **Install** beside **LFS SRE Tools** (2). It is switched on straight away.

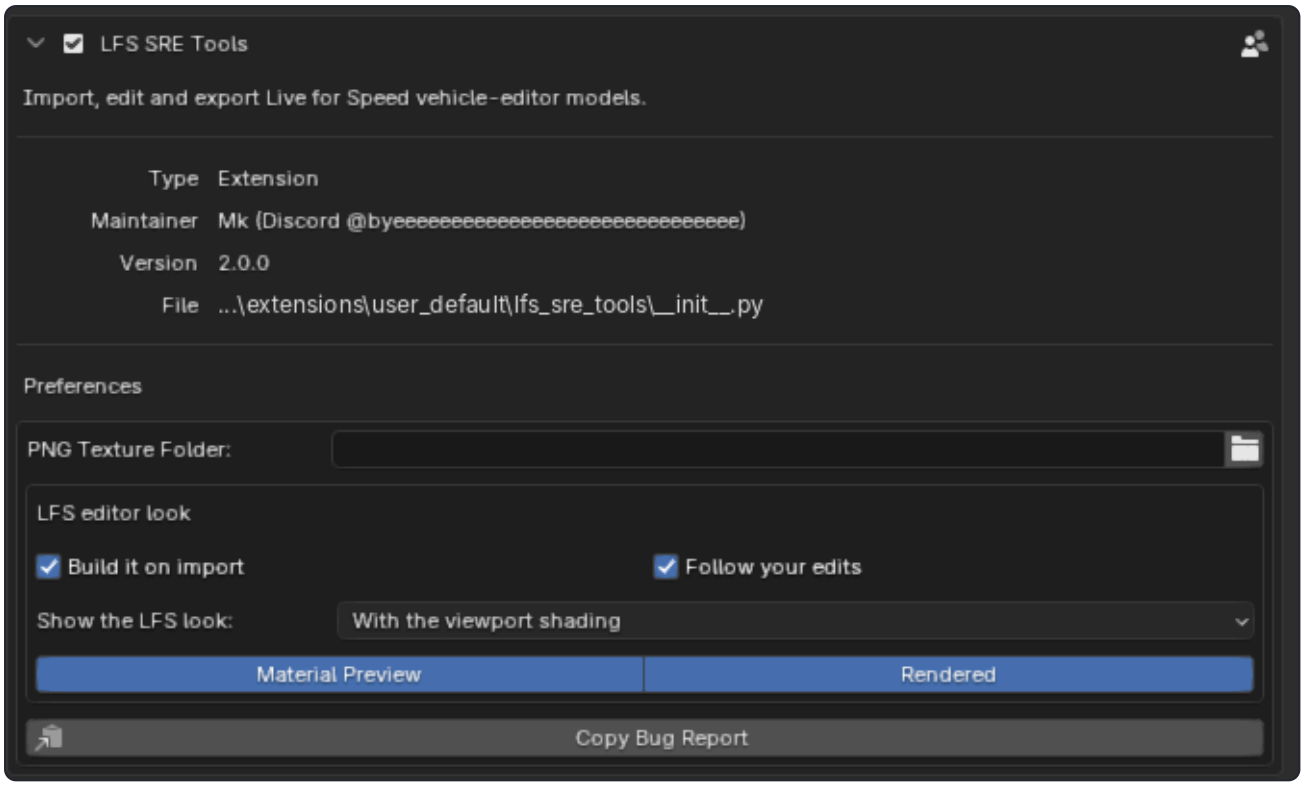


7 Close the Preferences. With the mouse over the 3D View press `N` and click the **LFS** tab 1 .



Without updates: drag `lfs_sre_tools-2.0.6.zip` onto the Blender window and press **Install**, or use **Install from Disk...** in the small arrow menu at the top right of **Preferences › Add-ons** . Keep the zip zipped. A copy installed this way never updates itself.

Search for *LFS* in the Add-ons list and click the arrow beside **LFS SRE Tools** to see its settings.



`Fig. 2` The add-on in Preferences, expanded.

Its settings:

PNG Texture Folder

Your LFS texture folder (`data\png`). For imported cars you can leave it empty: the importer finds it next to the `data\3dob` folder the `.sre` is in. Set it if you start from FBX or OBJ models, because Bake Textures for LFS writes its pages there.

Check for updates when Blender starts

Shown when the add-on came from the repository. It is Blender's own *Check for Updates on Startup* box for that repository: ticked, the status bar says *Extension Update Available* when a new version is out.

Build it on import

Builds the LFS look when a car is imported. A second Blender does that in the background for about half a minute, so switch it off on a slower PC: imports are quicker, and **Build LFS look** in the SRE Import panel makes the look when you want it.

Follow your edits

When you switch to Material Preview, the look first catches up with your edits. Off on a slower PC: the look stays as it is until you press **Rebuild LFS look**.

Show the LFS look

With the viewport shading (the default): Solid and Wireframe show the model you edit, the modes switched on below it show the LFS look. *As a separate scene*: the look gets its own scene and the LFS tab switches between the two.

Copy Bug Report

Copies the versions, these settings, the scene in short and the last messages, ready to paste where you ask for help. File paths are left out, so your user name stays private.

UPGRADING If you used the older single-file versions (*LFS SRE Exporter*, *LFS SRE Importer*), untick and remove them first. Two copies of the same tools register the same buttons and get in each other's way; the LFS tab shows a red warning while both are on.

SET IT UP ONCE

Getting updates

Installed from the update link (Install)? Then you are set: Blender looks for a new version when it starts and shows an **Update** button in Get Extensions. To look right away, click the small arrow at the far top right of Get Extensions and choose **Refresh Remote**. The update link is `https://byemk.github.io/lfs-sre-tools/index.json`.

1. A copy you installed from the zip does not update. Remove it in **Preferences › Add-ons** first.
2. Then add the update link as in Install and install **LFS SRE Tools** from Get Extensions.

From then on Blender tells you when there is a new version, and **Update** installs it. To look for updates Blender needs *Allow Online Access* (**Preferences › System**).

SHORTCUT The release page, `https://byemk.github.io/lfs-sre-tools/`, has an *lfs_sre_tools* link: dragged onto Blender it adds the repository and installs in one go. That does not work from every browser; if nothing happens, use the steps above.

CHECK THIS ON EVERY PC

Folders on your PC

The add-on needs to know one folder: your LFS **texture folder**, `data\png` inside your LFS folder (for example `C:\LFS\data\png`). On your PC it may be found by itself; on someone else's it may not, so check it once.

FOLDER	WHERE IT IS	WHAT TO DO
Texture folder	`data\png` in your LFS folder	Found by itself when the `.sre` is in `data\3dob`, or when the car's `.png` files sit next to the `.sre`. Anywhere else, set **PNG Texture Folder** in the add-on's preferences or in the box in the SRE Import panel. Blender saves it with its preferences. Once a folder has been found, later imports in the same `.blend` use it too. With nothing found, it also looks in the usual LFS places (`C:\LFS` and the like on each drive).

FOLDER	WHERE IT IS	WHAT TO DO
Cars	`data\3dob` in your LFS folder	Import from anywhere. Export into `data\3dob` so the vehicle editor lists the car.
Baked pages	the texture folder	**Bake Textures for LFS** writes its pages there, so set the folder before baking a model from FBX or OBJ.

If textures are missing after an import, the status bar says how many texture pages were not found and which folder was searched. The SRE Import panel in the LFS tab then shows a box to pick the folder. Pick it and import again.

`NOT MISSING` Pages called `s_mirr`, `s_clock` or `s_plate` are drawn by LFS itself (mirror, clock, number plate). They are not files and are not counted.

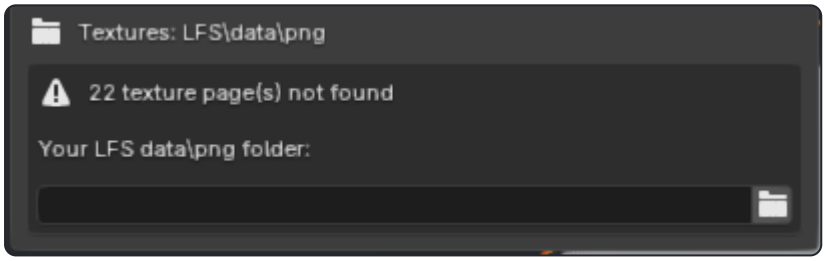


Fig. 3 A car imported from a folder with no textures nearby: the panel asks for the texture folder.

STEP 1

Import a car

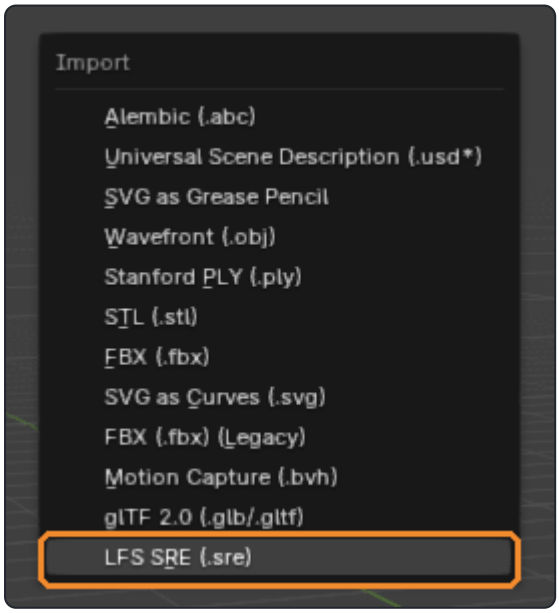


Fig. 4 The importer sits with Blender's own.

Choose **File › Import › LFS SRE (.sre)** and pick a car from your LFS `data\3dob` folder. The SRE Import panel in the LFS tab has the same button.

Each sub-object of the file becomes one Blender object, named as in the editor. The textures load from `data\png`. The LFS look is built in the background; Blender stays usable while it builds, and the SRE Import panel shows *LFS look: building…* until it is ready.

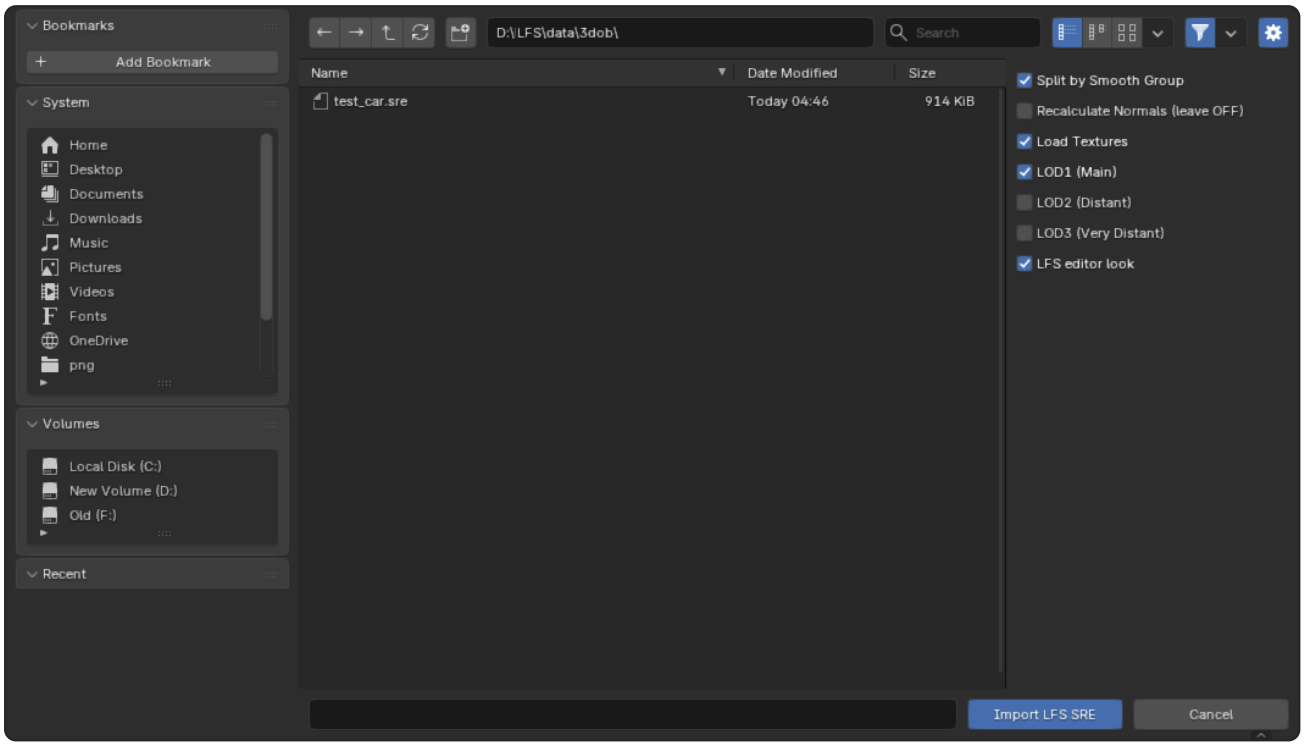


Fig. 5 The import window. The options are on the right; the defaults suit almost every car.

IMPORT OPTION	DEFAULT	WHAT IT DOES
LFS editor look	On	Builds the display-only copy lit like the vehicle editor (next section).
Load Textures	On	Loads the texture pages the car uses from the texture folder.
LOD1 (Main)	On	The detailed faces you see up close. This is what you edit.
LOD2, LOD3	Off	The low-detail faces LFS shows far away. They are kept in the file either way.
Recalculate Normals	Off	Leave it off. Interiors and wheels face inward on purpose.
Split by Smooth Group	On	Only for older `.sre` files that did not come from the vehicle editor.

STEP 2

The look and the model

An imported car has two faces in Blender, and the viewport shading buttons (top right of the 3D View, or `Z`) switch between them.



Fig. 6 Material Preview and Rendered show the **LFS look**, a copy you cannot edit.

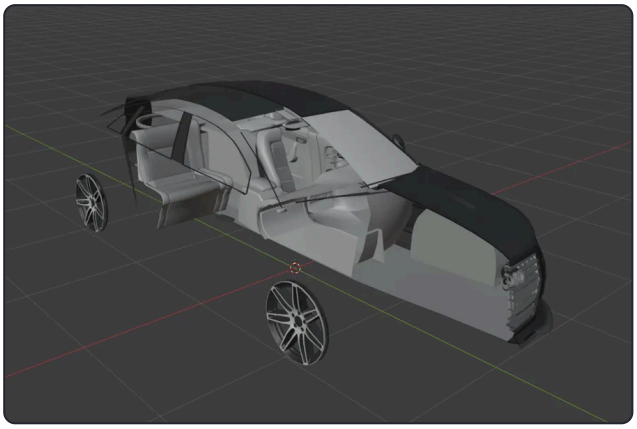


Fig. 7 Solid and Wireframe show the **model you edit** and export.

- **The look follows your edits.** Change the mesh in Solid, switch to Material Preview, and the look is rebuilt from what an export would write at that moment. You see the car the way the editor will show it before you export anything.
- **The mirrored side.** Faces marked as mirrored in the `.sre` exist once in Blender. LFS draws their mirror image on the other side of the car, and so does the look. An import shows that side in Solid view too (import option *Show mirrored side*), also while you model: an edit on one side shows on the other at once. It is only for the eye: never exported and not selectable. **Show Mirrored Side** in the LFS SRE panel turns it off and on.
- **The lighting** is in the LFS Look panel of the LFS tab: sun turn, height, strength and colour, shadow fill, sky reflections and the two paint colours. The defaults are measured from the LFS editor.
- Blender compiles shaders before it shows the look for the first time, so give it a few seconds. Switching back and forth is quick after that.
- **On a slower PC** switch off *Build it on import* and *Follow your edits* in the add-on's preferences. Build or rebuild the look with the button in SRE Import when you want to see it.

WHERE THE TOOLS LIVE

The LFS tab

Press `N` over the 3D View to open the sidebar, then pick the **LFS** tab on its right edge. Everything except import and export lives here.

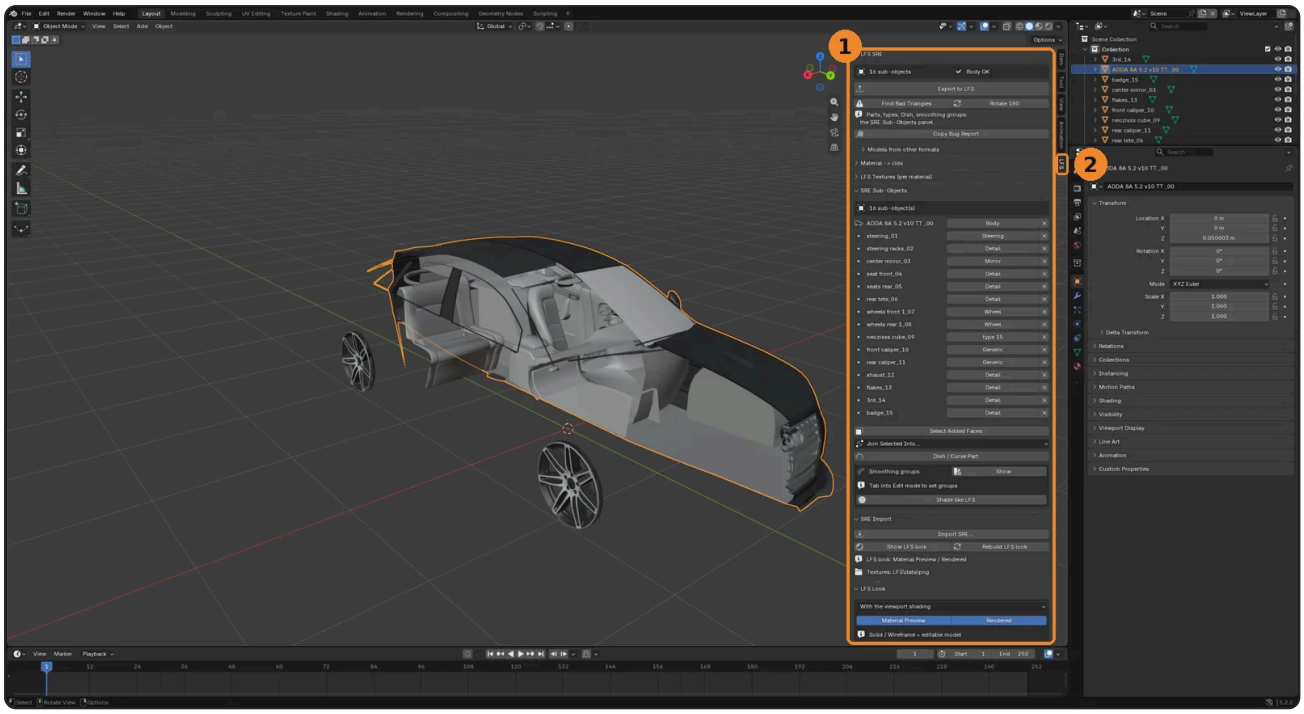


Fig. 8 The sidebar (1) with the LFS tab picked (2). Drag the sidebar's left edge to make it wider.

LFS SRE

The sub-object count with a body check, **Export to LFS**, Find Bad Triangles, Rotate 180, Show Mirrored Side and Copy Bug Report. For a model from another format also the numbered steps **1 Auto Setup** and **2 Bake Textures for LFS** with Undo Bake, a red note when the car faces backwards, and the closed *Models from other formats* section with its tools.

LFS Map per Material

Which LFS map (the editor's map number) each material uses; imported materials already have theirs.

LFS Texture per Material

Which texture each material gets in LFS.

SRE Sub-Objects

The car's parts and their types, Select Added Faces, Join Selected Into..., Dish / Curve Part and the smoothing-group tools.

SRE Import

Import, switch between the look and the editable model, rebuild the look, and the texture folder in use.

LFS Look

The look's lighting and where it shows.

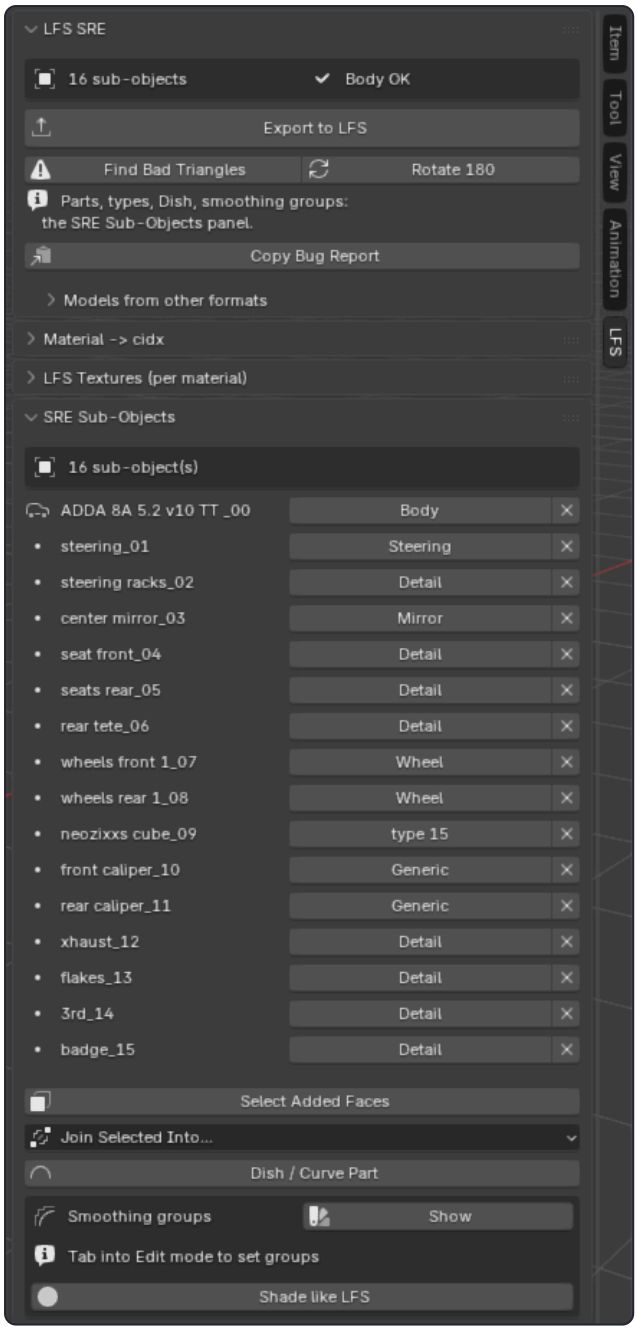


Fig. 9 Top of the tab.

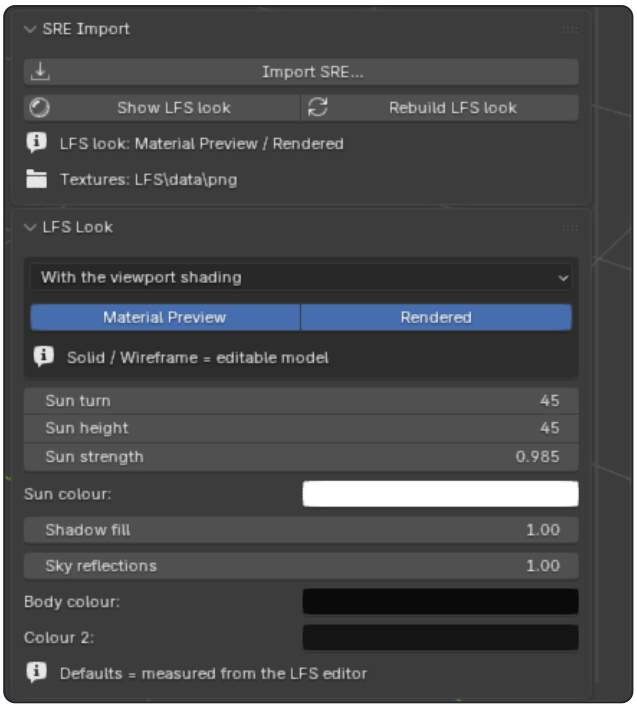


Fig. 10 Scrolled down.

STEP 3

Edit and export

The imported objects are ordinary Blender meshes. Move points, extrude, cut, delete, add faces, swap materials. Work in Solid view and look at the result in Material Preview.

To export, use `File › Export › LFS SRE (.sre)` or **Export to LFS** in the LFS tab, then open the file in the LFS vehicle editor.

- A sub-object you did not touch is written back **byte for byte**.
- In a sub-object you edited, every face that is still where it was keeps its map, mirror flag, LOD and smoothing group from the file. New and changed faces take them from their material and their neighbours.

- Export an unchanged import and you get the same file back. That makes a quick test that everything is in order.
- The export window starts where the car went last, else where it came from, with its name. Writing over an `.sre` keeps the old file as `name.sre.bak`; to go back, rename it to `.sre`.
- The status bar says when the car uses texture pages your `data\png` does not have (the editor shows them blank), and when the file is saved outside a `data\3dob` folder (the editor lists cars from there). A progress number on the cursor shows a big export is still working.

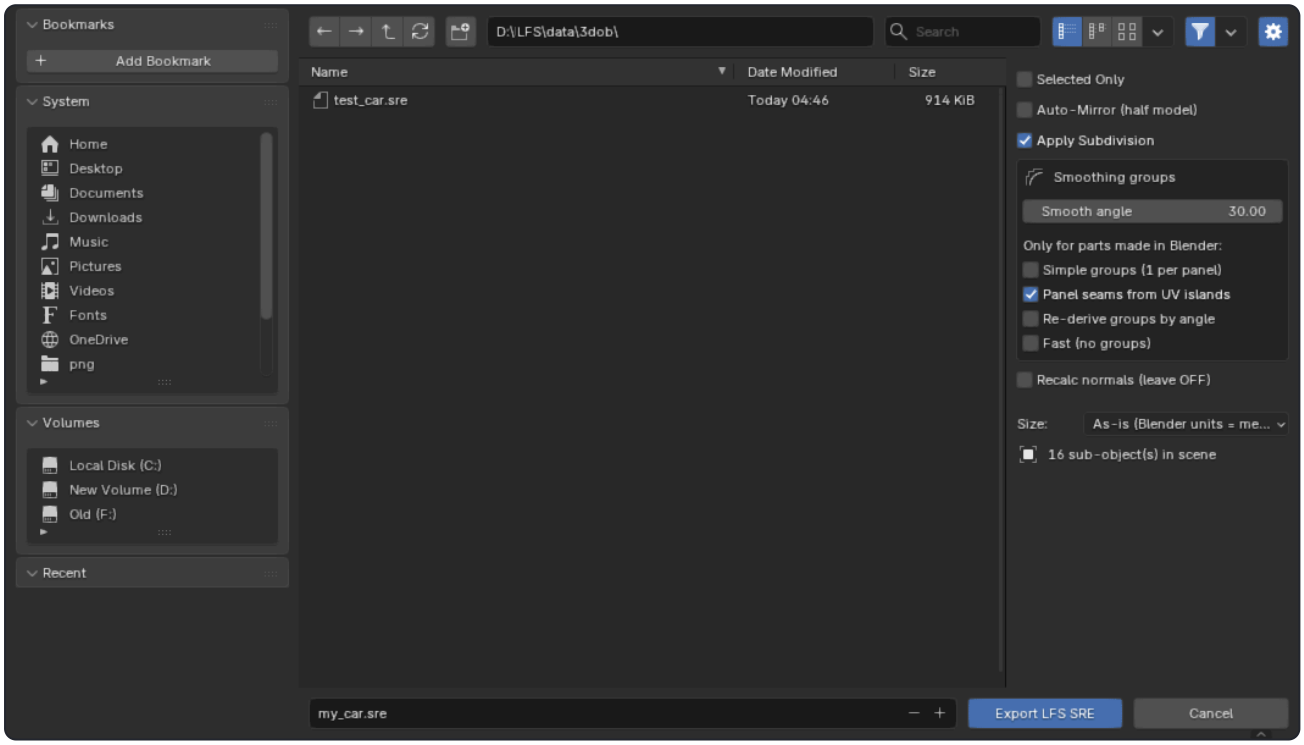


Fig. 11 The export window with its options. Save into `data\3dob` so the vehicle editor finds the file. Most of the Smoothing groups box only matters for parts made in Blender.

EXPORT OPTION	DEFAULT	WHAT IT DOES
Selected Only	Off	Export only the selected sub-objects.
Auto-Mirror (half model)	Off	For a mesh that is half a car: LFS mirrors it. Leave it off for full models.
Apply Subdivision	On	Bakes Subdivision Surface modifiers into the export. Turn it off if a part goes over the 32767-point limit.
Smooth angle	30°	Faces that meet at more than this angle get a hard edge. It decides which new or edited faces of an imported part shade into the faces around them. The window starts from the angle the scene uses.

EXPORT OPTION	DEFAULT	WHAT IT DOES
Simple groups (1 per panel)	Off	Parts made in Blender only: one group per panel, worked out from the UV islands.
Panel seams from UV islands	On	Parts made in Blender only: a hard edge along the borders of UV islands, for game models whose panel gaps are only painted on.
Re-derive groups by angle	Off	Parts made in Blender only: hard edges by the smooth angle as well as the sharp edges you marked. A part with no sharp edges marked is split by the angle anyway.
Fast (no groups)	Off	Parts made in Blender only: no smoothing groups at all. Quick, for scenery.
Recalc normals	Off	Leave it off; it can flip interiors and wheels.
Size	As-is	Blender units are metres. *Fit to real length* scales the model so its longest side matches a length you type.
Export past LFS's limits	Off	Writes a part past 236 maps or 238 cutouts anyway (the editor crashes on those). The format's own 32767 points and 65535 triangles per part always apply.

TIP Save to a new file name and check it in the vehicle editor before you overwrite a car you care about.

HOW A CAR IS BUILT

Sub-objects and new parts

An `.sre` car is a list of sub-objects: one body plus parts such as wheels, the steering wheel, the mirror and the interior. In Blender each one is an object, and SRE Sub-Objects lists them with their type.

TYPE	SHOWN AS	USED FOR
0	Body	The main body. A car has exactly one; **Export to LFS** stays grey until it does.

TYPE	SHOWN AS	USED FOR
3	Detail	Most parts: interior, trim, details. Auto Setup also uses it for wheels.
2	Steering	The steering wheel.
10	Mirror	The interior rear-view mirror.
1	Generic	A generic part.
12	Wheel	The road wheels in cars made in the vehicle editor.
other	type N	Kept exactly as the file has it.

- Click a type to change it. A small dialog opens with *Keep as it is* selected, so nothing changes by accident. The `×` beside it unmarks the part.
- A part file (a single wheel or an interior `.sre`) has no body. That is fine: export it with **File › Export**.
- **Select Added Faces** selects everything you modelled since the import in the active part. Handy on triangulated meshes.

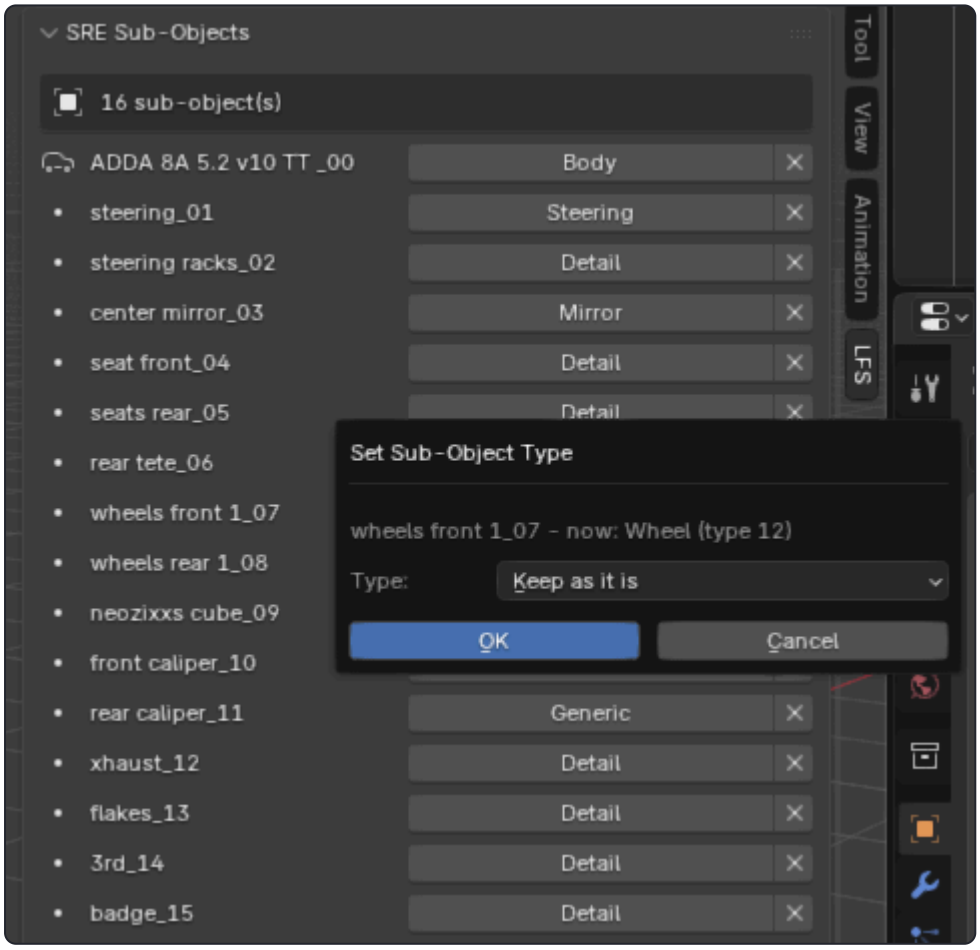
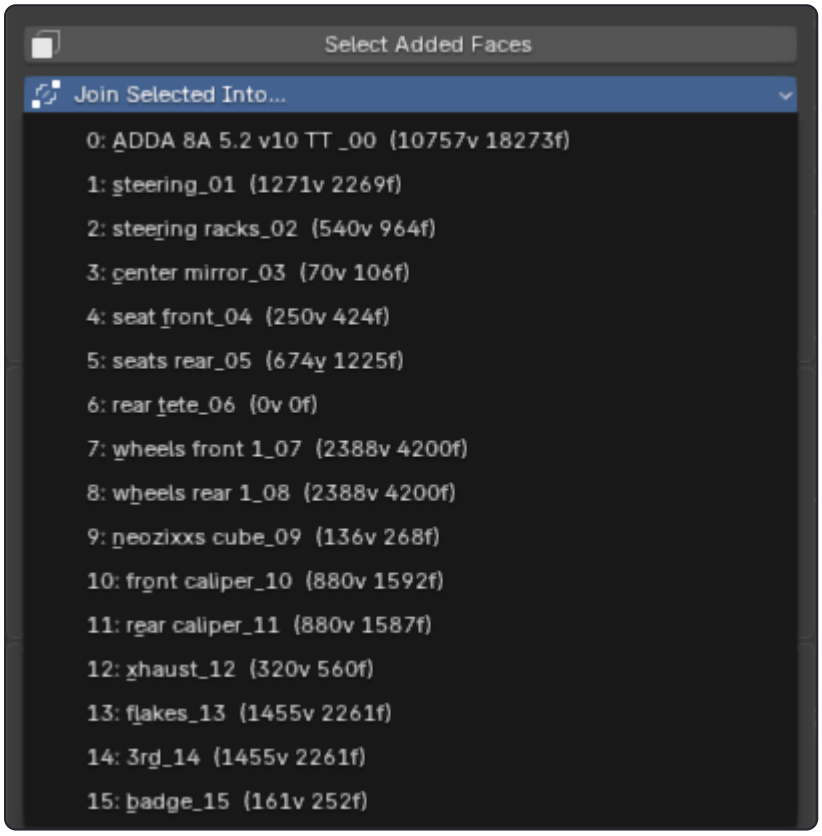


Fig. 12 Changing a part's type.

Add a part to an imported car

1. Model the part, or import or append it, in Solid view.
2. Give it materials. One of the car's own materials (paint, chrome, glass) makes it look like that part. A new material exports as its plain colour.
3. Select it, open **Join Selected Into...** in SRE Sub-Objects and pick the part it belongs to, such as the body for a spoiler.
4. Export. The new faces get smoothing groups by themselves (see below).



`Fig. 13` **Join Selected Into...** lists the car's parts.

`WHY JOIN` **Mark Selected Part(s)** makes a separate new sub-object instead. On big cars it can stop with *Map table too long*, because the editor crashes past about 236 maps. A joined part uses the maps of the part it joins.

`SHAPING TOOL`

Dish / Curve Part

Curves a part in or out with a smooth falloff: a dished wheel face, a domed panel, a concave diffuser, a flared lip. The centre, the axis and the radius come from the part itself, so nothing has to be dragged by hand.

1. Select the part. In Object mode it works on the whole object; in Edit mode on the selected faces.
2. Click **Dish / Curve Part** in SRE Sub-Objects.
3. Open the panel at the bottom left of the 3D View (or press `F9`) and set **Depth**. Negative pulls the centre in, positive pushes it out. The first time, Depth is 0 and nothing moves until you set it; after that the button repeats your last settings.

WHEELS For a concave wheel, set **Zone** to *Ring*: the spokes bow in while the hub and the rim's edge stay where they are. *Centre* moves the hub as well, and it comes out of the back of the rim (from the front it looks like a second rim behind the first).



Fig. 14 The front wheel as imported.



Fig. 15 After Dish / Curve Part: the *Deep dish* curve, Depth −0.05 m, from 0.12 to 0.85 of the radius.

SETTING	WHAT IT DOES
Shape	Smooth, Round, Cone, Bell, Power curve, or Custom curve (draw it yourself).
Depth (m)	How far the centre moves. Negative = in (a dish), positive = out (a dome).
Start / End radius	Where the curve starts and stops, as a share of the radius. 0.3 keeps the inner 30% flat, like a centre cap.
Taper (m)	Also moves points sideways: negative narrows the middle, positive flares it.

SETTING	WHAT IT DOES
Zone	Strongest at the centre, at the outer edge (a deep-dish lip) or halfway out (a ring).
Axis	*Auto* uses the direction the part is thinnest in: the axle on a wheel.
Centre	The selection centre, the 3D cursor or the object origin.
From original shape	On: re-applied from the shape before the first dish, so a smaller Depth makes it shallower again. Off: adds on top, to stack a dish and a lip.

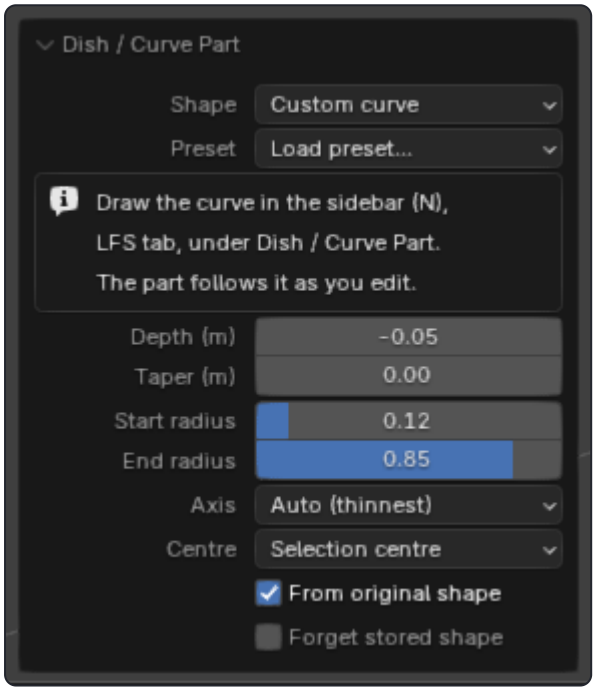


Fig. 16 The settings panel.

Draw your own curve

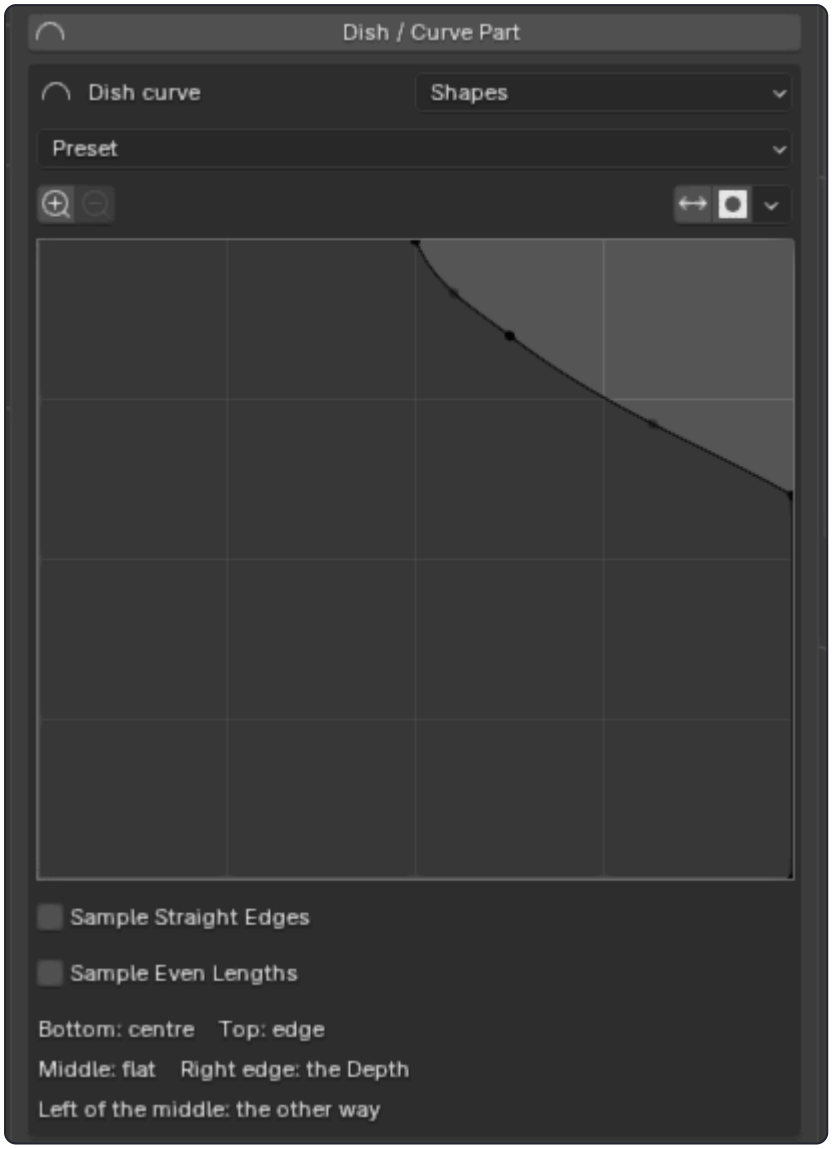


Fig. 17 The curve under the Dish button.

Set **Shape** to *Custom curve* and the curve appears in the sidebar, under the Dish button.

- **Bottom** is the centre of the part (Start radius), **top** is its edge (End radius).
- The line starts straight up the **middle**: flat, nothing moves.
- Bend it to the **right** and that ring moves by the Depth. The right edge is the whole Depth (it starts at 5 cm; change it in the settings panel). Left of the middle moves the other way.
- Drag the points; `Ctrl` + click adds one. In Object mode the part follows while you drag, in Solid and in Material Preview; in Edit mode it follows when you let go.
- **Shapes** loads a starting curve: flat, linear, smooth, round, bell, deep dish, lip or ring.



Fig. 18 The dished wheel in Material Preview. The look follows the edit.

SHADING

Smoothing groups

LFS shades by smoothing group, a number from 0 to 23 on every face (LFS has 24). Faces of one group that share a point blend smoothly; a different group makes a hard edge. Each face also has a contribution level (overridden, default, medium, high), and at each point only the faces with the highest level count.

What an export writes

- A face that is still where it was in the file keeps its group.
- A face you added or changed joins the group of a neighbour across a smooth edge: not marked sharp, under the smooth angle (*Smooth angle* in the export window, 30° by default), and shaded smooth.
- Otherwise it gets a group that no face sharing a point with it uses. Extrusions come out crisp; a part welded on flush shades into the panel.
- A group you set by hand always wins.
- Every Blender face keeps one group, so a quad never gets a crease across its diagonal, and new groups stay within LFS's 0 to 23.

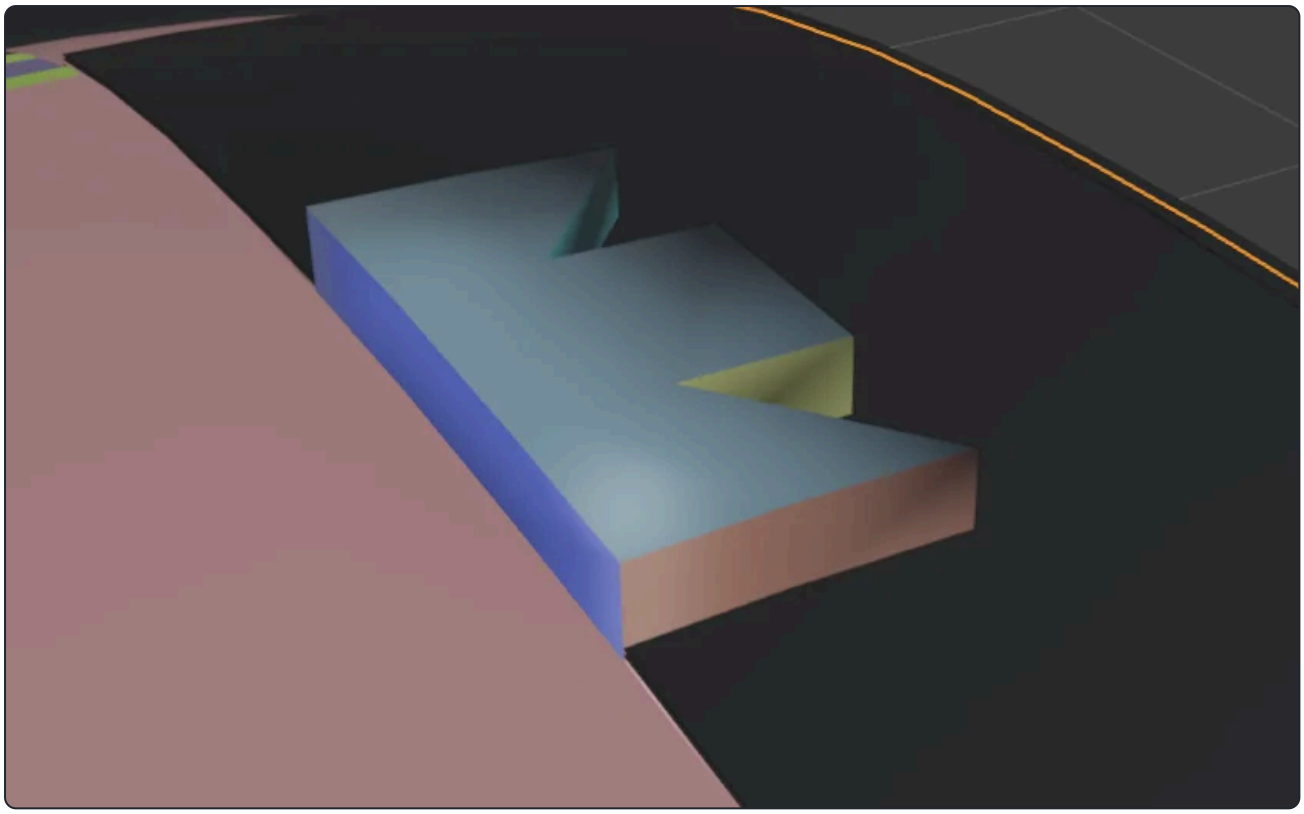
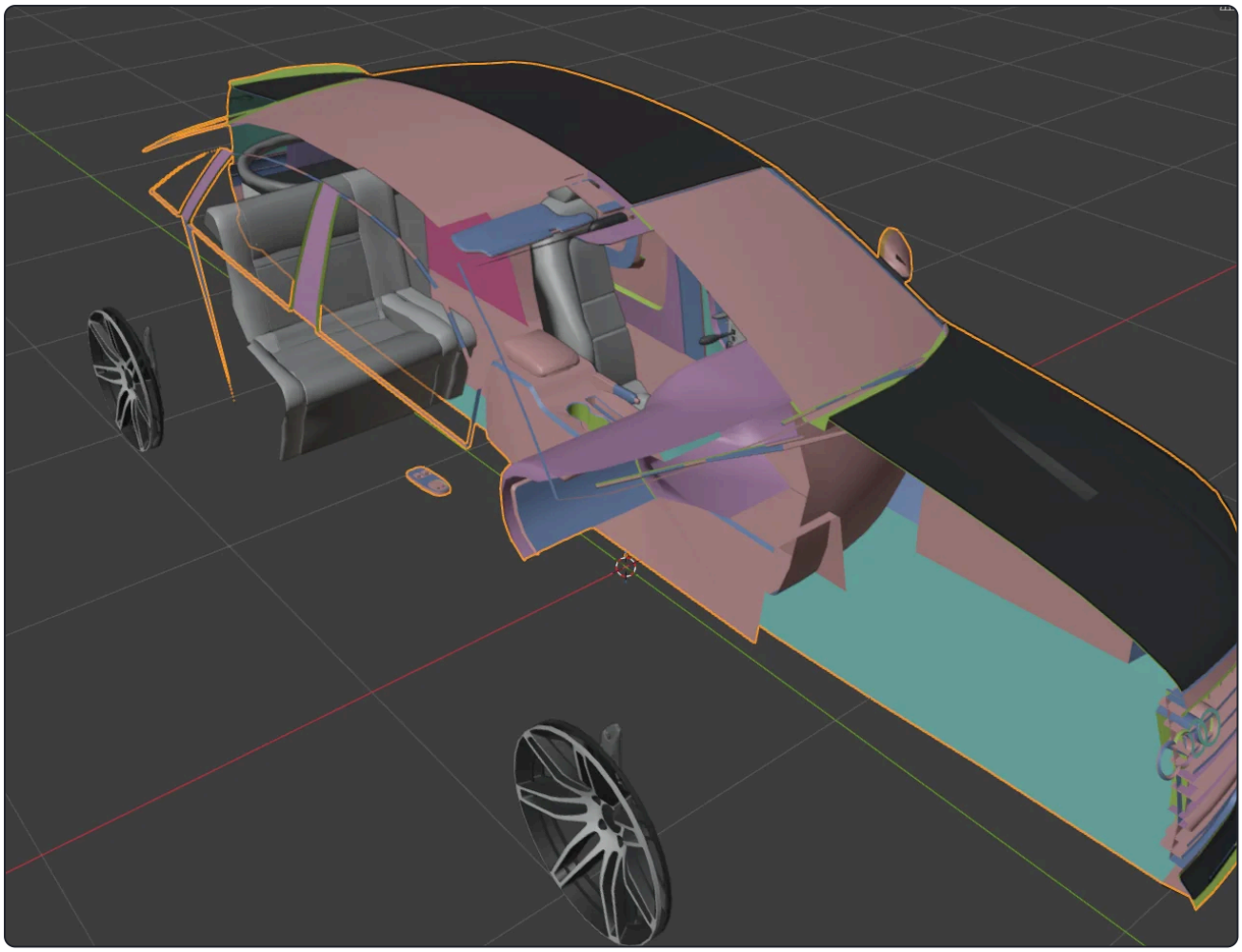


Fig. 19 A roof panel extruded upward, coloured by group: the raised cap and each new wall get groups of their own (crisp edges), and the roof around keeps the group it had in the file.



`Fig. 20` **Show** colours the active part by the group each face will get, new and edited faces included. Press it again to hide.

New group

A group nothing else in the part uses: smooth inside the selection, a hard edge all round.

Join around

The group of the faces around the selection, so it shades into them.

Flat

A group of its own for every face: faceted.

Automatic

Forget what was set by hand and let the export decide.

Number...

Type a group number from 0 to 23, like in the editor.

Select group

Selects every face in the same group as the selected ones.

Shade like LFS

Sets Blender's normals the way LFS will shade the part, worked out from the triangles the export writes, so you see the result before exporting. It marks the edges between different groups sharp, as the importer does, so Blender does not blend across them, and leaves flat and smooth faces as they are.

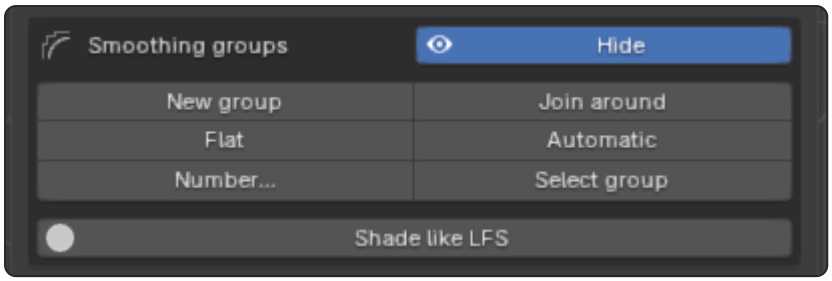


Fig. 21 In Edit mode the box adds the tools for the selected faces.

FBX, OBJ, GLTF

Cars from other formats

For a model that did not come from an `.sre`: import it into Blender or open its `.blend` (Blender units are metres; the export can also fit the car to its real length). Then the three numbered steps at the top of the LFS SRE panel:

1. **Auto Setup** sorts the meshes by name and material into body, interior, wheels and steering wheel, joins each group and marks it as an LFS part. It takes what the view shows: meshes that are hidden or in switched-off collections are left out, and of several levels of detail (LOD0...LOD3, Hires / Lowres) only the most detailed. It also finds textures that are not where the `.blend` says (pink in Material Preview), turns a car that faces backwards round (LFS's forward is +Y – by its steering wheel, by its head and tail lights or its wheels' names (FL, FR / RL, RR), by small red tail lights, or by the cab windows of a truck or a van; names at both ends win over a steering wheel near the middle, as on a tractor), stands up a car that came in on its nose or tail or lying along X (a file whose axes Blender did not convert), scales a car that came in centimetres, millimetres or inches to its real size, makes materials that look the same into one when a model has hundreds of them, gives parts that have no material one by their names (glass, tyres, rims, carbon, chrome... – the *auto* materials), and leaves out a floor, a studio panel or the scenery of a diorama (terrain, trees). The parts end up at the top of the Outliner; a 3ds Max car's empties go into a switched-off collection, *Auto Setup leftovers*. A Max copy that came over as an empty with nothing in it (a right door, a wheel) gets the pieces it copies, and when the car's outside is only one half of it (Max mirrored it and the mirror did not come over), LFS mirrors that half, with its centre line exactly on the car's middle. Check that the LFS SRE panel says *Body OK*.

2. **Bake Textures for LFS** makes the model's own textures work in LFS: they go into your LFS `data\png` folder as texture pages, and every surface gets an LFS map onto them. Until that folder is known, the LFS SRE panel shows a box to pick it. The dialog shows the options that matter; the rest are under *More options*. **Undo Bake** puts the model's own materials back.

3. **Export to LFS** and open the car in the vehicle editor.

- How the bake treats the model: flat pieces of its texture layout keep the model's own texture pixel for pixel (the textures are copied into `data\png`, 1024 at most). Round pieces whose texture is just a colour or a grain become a plain colour, inside the car LFS's leather tinted with their colour. Round faces with real detail (dials, labels, knobs, vents) are painted into a new texture page. Each piece is painted from as few views as it needs: a lamp, a badge, a screen or a dial is one whole picture, a seat two or three. The pixels follow the detail (dials, buttons and lamps up to three times, plain plastic a third), and inside the car also how well the driver sees a piece, so the gauges and the climate panel come out big and the floor mats small. The steering wheel is one picture seen from the driver (*More options: Steering wheel in detail*). *Method: Paint new texture pages* paints everything instead: softer, but every surface keeps its texture. LFS has no normal maps: their bumps (seams, grille slats, vents, dial rings, an embossed badge) are drawn into the colour as thin lines (*More options*).
- **Car paint** (Bake dialog) decides what happens to the paint's own texture. *Into a skin* (the default) paints it into an LFS skin, `data\skins\<NAME>_DEFAULT.jpg` (1024, the size LFS World asks for, or 2048 under *Skin size*), seen from the top, both sides, the front and the back like LFS's own cars. Only the outside goes into the skin: paint on interior parts, or hidden behind other parts (door jambs, the inside of the doors, what is behind the windows), keeps the paint's own colour. The paint is the material that covers most of the outside. The body stays skinnable, and the skin is not part of the 5 MP texture budget. Colour 1 still tints the paint: white shows the skin as it is. Pick the skin and the colours in the vehicle editor's *Colours* tab. LFS names a skin `<Skin ID>_<NAME>`, and a mod gets its Skin ID when you upload it. Then rename the file and the skin page in the editor, or bake again with *Skin name* set. *Into texture pages* bakes the paint like any other part (it uses the budget and the body is no longer skinnable). *Plain paint colour* keeps one LFS paint colour.
- A car that still faces backwards (the LFS SRE panel says *Faces backwards for LFS*): **Rotate 180** turns every part.
- The tools for these models are in the closed **Models from other formats** section of the LFS SRE panel: Mark Selected Part(s) for a mesh Auto Setup missed, Prep Steering and Spin for a new steering wheel, the mesh repairs and the texture-map tools.
- LFS draws each face from one side only. **Fix Inside-Out Shells** turns shells whose faces point inward; **Mark Double-Sided** makes thin sheets such as seat backs and glass visible from both sides. Mark Double-Sided doubles every sheet it finds, so check the triangle count afterwards.
- **Keep my texture layout** (Bake dialog): for textures you made for LFS yourself. The model's own textures go to LFS laid out exactly as they are: nothing is cut out and rearranged, painted into new pages, shaded with bumps or swapped for LFS's leather. Round pieces keep their texture too (it may stretch a little). Only see-through parts (glass, lamp lenses) get a page of their own, because LFS needs that. A texture over 1024 px goes to LFS at 1024.
- **LFS Texture for Selected Faces** (LFS tab, in Edit mode): select faces (the seats, a door card, the dash top) and give them LFS leather, LFS carpet, a plain colour, Bake or Bake

sharper, or Automatic to undo. Run Bake Textures for LFS again afterwards.

- **A texture page for each part** (Bake dialog, on by default): each part – body, wheels, interior, steering wheel – gets texture pages of its own, named after it (`<name>_bk0body.png`, `<name>_bk1int.png` ...), each as big as its pieces need, instead of all parts sharing the same pages. Together they stay inside the 5 MP budget. Handy when you want to repaint or swap one part's texture on its own.
- Bake never writes over a page that is already in `data\png`. A model texture with the same name as one of the editor's pages (a `carpet.png`, say) gets a name of its own, such as `carpet_2`, and each `.blend` keeps its own names for its baked pages.
- Bake is for models from other formats. On a car imported from an `.sre` it says so and changes nothing: give a new part one of the car's materials or a plain colour.

MODEL BY MODEL

Baking different kinds of models

Bake Textures for LFS turns a model's own textures into what LFS can draw: square texture pages in `data\png` (1024 px is LFS's upload limit; 2048 and 4096 work in the editor only), 5 MP for the whole car, and an LFS map for every surface. LFS lays a texture on flat from one direction only. So a flat piece can keep the model's own texture as it is, while a round one either stretches it, gets a plain colour or is painted into a new page. Which is best depends on the model. Find yours below, then check the result (below).

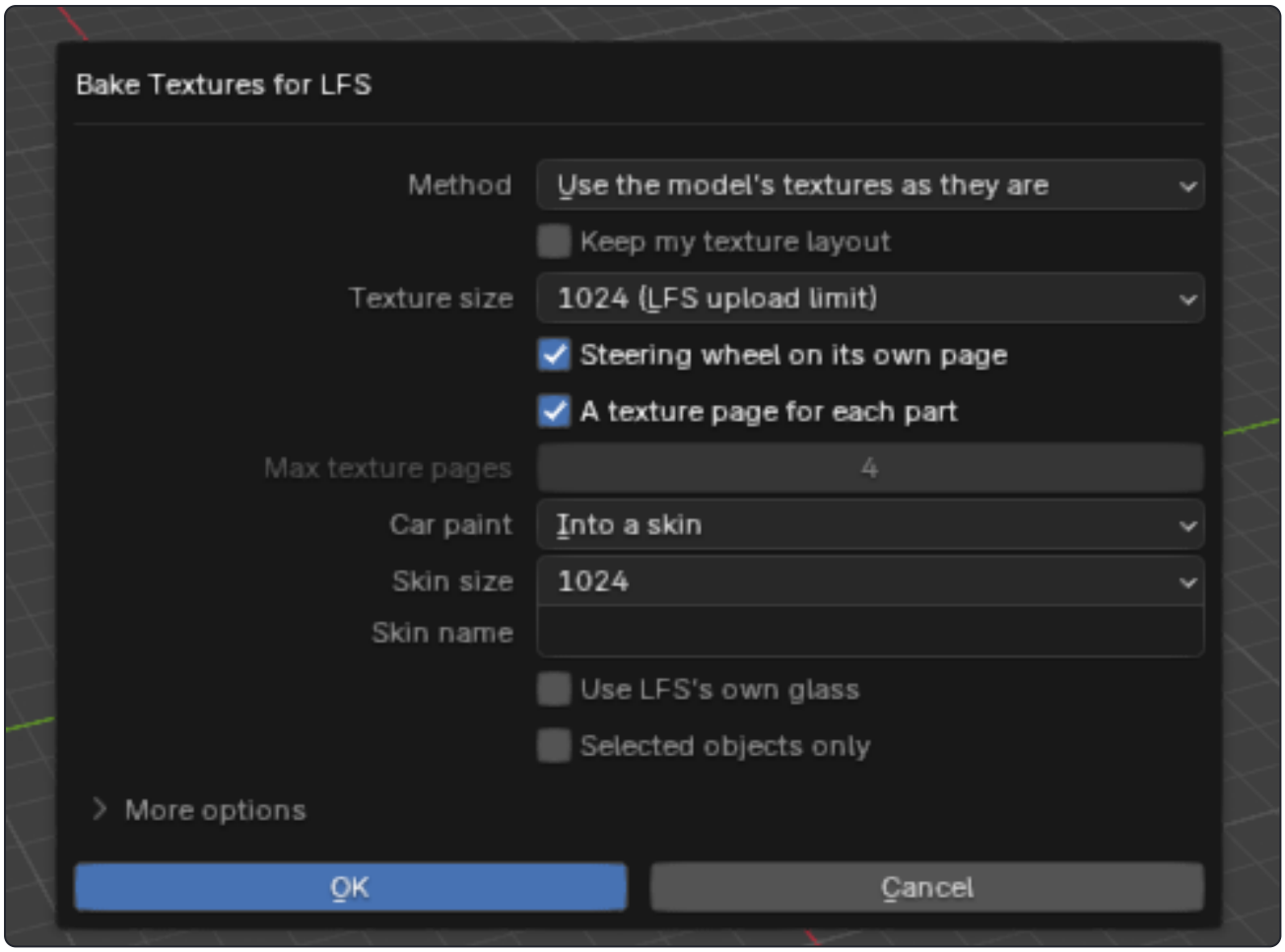


Fig. 22 The Bake dialog: *Method* and *Keep my texture layout* at the top decide most of it. The rest is under *More options*.

YOUR MODEL	WHAT TO PICK	WHAT YOU GET
A downloaded car with its own textures (an FBX or `.blend` from a model site, a game rip)	The defaults: *Method: Use the model's textures as they are*, *Round parts: Automatic*.	Flat pieces (door cards, console, dash, badges) keep the texture pixel for pixel. Round pieces with detail are painted, each from as few views as it needs: a lamp, a badge or a dial is one picture, a seat two or three. Inside, the pixels go to what has detail and what the driver sees (gauges, climate panel), and the steering wheel is one picture seen from the driver.
Textures you made for LFS yourself (your own pages, laid out the way you want)	Tick *Keep my texture layout*.	Your pages go to LFS as they are: nothing is cut out, rearranged, painted over, shaded or swapped for LFS's leather. Round pieces keep your texture too, though it may stretch a little. Only see-through parts (glass, lamp lenses) get a page of their own. Pages go in at the *Texture size* at most.

YOUR MODEL	WHAT TO PICK	WHAT YOU GET
A 3ds Max car converted to Blender	Just run Auto Setup and read its report, then bake as above.	Auto Setup puts the parts at the top of the Outliner and the Max empties into *Auto Setup leftovers* (Fig. 23). It fills Max copies that came over empty (a right door, a wheel). When the outside is only one half, LFS mirrors it, with the centre line exactly on the car's middle. It also scales a car in decimetres or inches to metres.
A model without picture textures (plain colours, procedural materials)	*Car paint: Plain paint colour*. Leave *Seats, leather, cloth: LFS leather* on.	Every material goes to LFS as its colour (Blender's procedural textures cannot go to LFS). Seats and cloth get LFS's leather tinted with their colour, so they don't look like plastic.
A model whose texture layout does not suit LFS (overlapping or very curved UVs: the result looks smeared or stretched)	*Method: Paint new texture pages*.	Everything is painted into new pages from the model's look. It works on any model and every surface keeps its texture, but it is softer than the model's own pixels.
A very detailed model (hundreds of thousands of faces)	Make a lighter version first (Decimate, or the model's low LOD).	LFS parts take at most 32 767 points each and LFS cars usually have under 100 000 faces. Auto Setup splits a part that is over the limit, and Bake takes much longer on a heavy model.

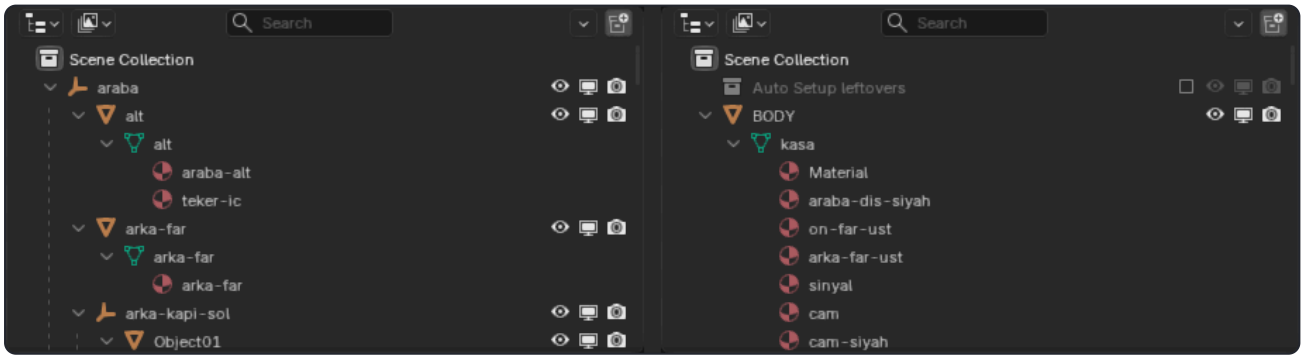


Fig. 23 A 3ds Max car in the Outliner before Auto Setup (left: its pieces hang under the Max empties) and after (right: the parts at the top, the empties in the switched-off *Auto Setup leftovers*).

The options, one by one

Method

Use the model's textures as they are (default) keeps the model's own texture on every flat piece and paints only what it has to. *Paint new texture pages* paints everything.

Keep my texture layout

For your own LFS textures: the model's pages go to LFS exactly as laid out (see the table).

Texture size

The largest page side. 1024 is LFS's upload limit. 2048 and 4096 are for the editor only.

Steering wheel on its own page

The wheel gets a page to itself: one picture of its front, seen from the driver.

A texture page for each part

On by default: the body, the wheels, the interior and the steering wheel get pages of their own (`<name>_bk0body.png` ...), handy for repainting one part. Off, all parts share pages (*Max texture pages*).

Car paint

Into a skin (default) paints the paint's texture into an LFS skin in `data\skins` and keeps the body skinnable. *Into texture pages* bakes it like any part. *Plain paint colour* keeps one LFS paint colour.

Use LFS's own glass

Windows get LFS's glass page instead of a baked see-through page.

Round parts (seats...)

Automatic: painted when their texture has detail, else a plain colour or LFS leather. *Plain colour*: the texture's average colour. *Paint into new texture pages*: always painted. *One map each (may stretch)*: the model's own texture, stretched over the curve.

Bumps from normal maps

LFS has no normal maps, so the bumps they show (vents, grille slats, seams, dial rings) are drawn into the colour. Off: the colour stays exactly the texture's.

Steering wheel in detail

The wheel's own texture (logo, stitching, buttons) is painted with extra pixels.

Seats, leather, cloth: LFS leather

Seat, leather and cloth materials get LFS's repeating leather tinted with their colour, so the page space goes to the dash and the badges.

Flatness needed

How flat a piece must be to keep its texture as it is. Higher keeps more pieces, with more stretching.

Directions

The views pieces are painted from: 6 (a box) is enough for most cars, 18 or 26 fit tilted pieces better.

Max cutouts

How many painted pieces one part may have (LFS's map list holds 236 per part). Fewer pieces means bigger merged ones.

Interior detail boost

How many times more pixels the inside of the car gets than the outside.

Texture name

The start of the page names. Empty: the `.blend`'s name. Bake never writes over a page that is not its own.

One texture for some faces

A part is one object, but you can still choose per face. In Edit mode, select faces (the seats, a door card, the dash top) and use **LFS Texture for Selected Faces** in the LFS tab: *LFS leather*, *LFS carpet*, *Plain colour*, *Bake*, *Bake sharper* (twice the pixels), or *Automatic* to undo. Then run Bake Textures for LFS again. For a whole material, use **LFS Texture per Material**.

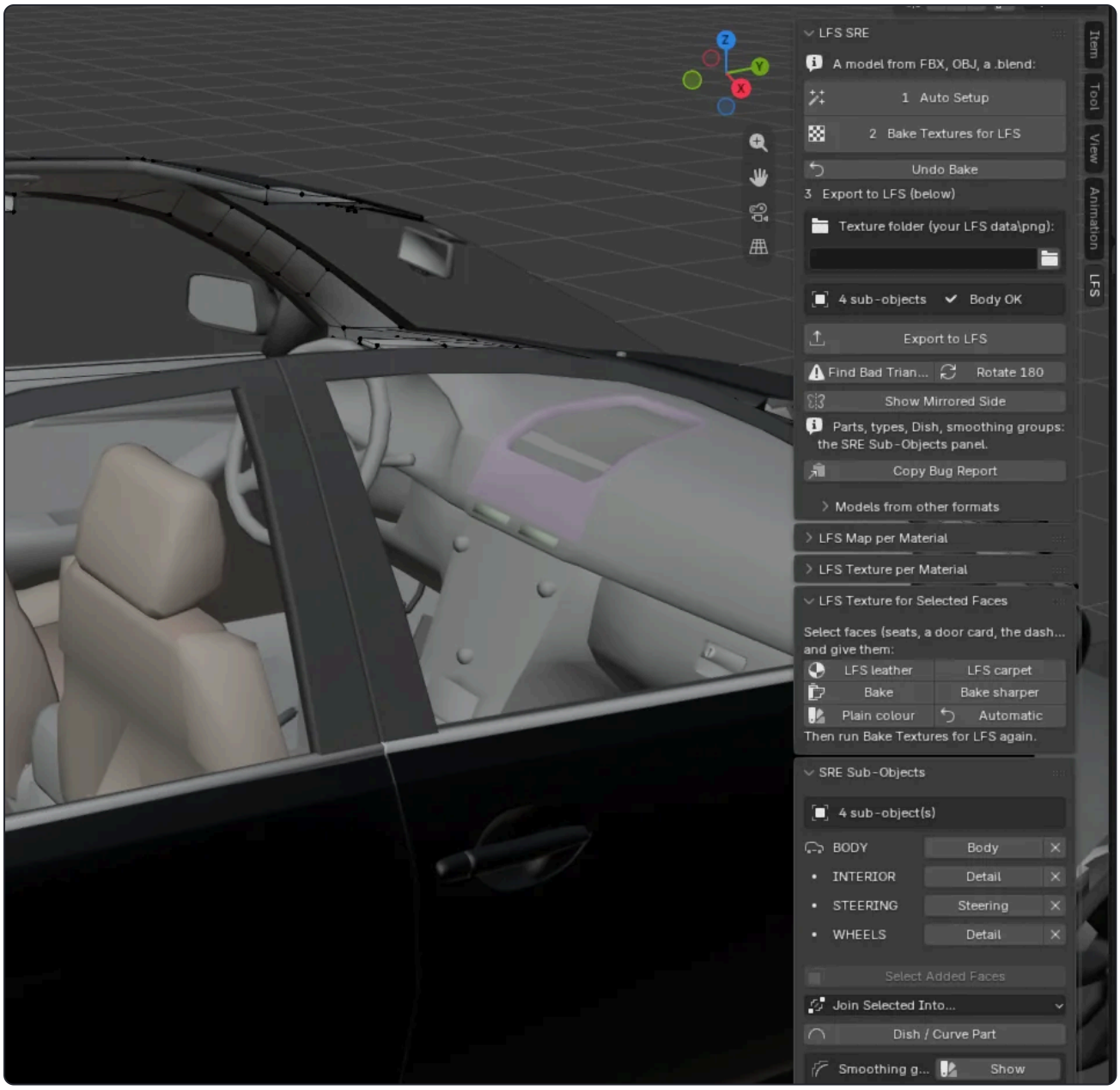


Fig. 24 The seats selected in Edit mode, and LFS Texture for Selected Faces in the LFS tab.

Check the result

- The report at the end of the bake says how many pages it made, how many pieces were painted and how many keep the model's own texture.
- Open the pages in `data\png` (`<name>_bk0body.png`, `<name>_bk1inter.png` ...): the details should be big and the plain parts small (Fig. 25).
- Material Preview shows the car the way LFS will draw it (the LFS look). Then **Export to LFS** and open the car in the vehicle editor.
- Not happy? **Undo Bake** puts the model's own materials back. Change the options and bake again: every bake starts from the model's own materials.

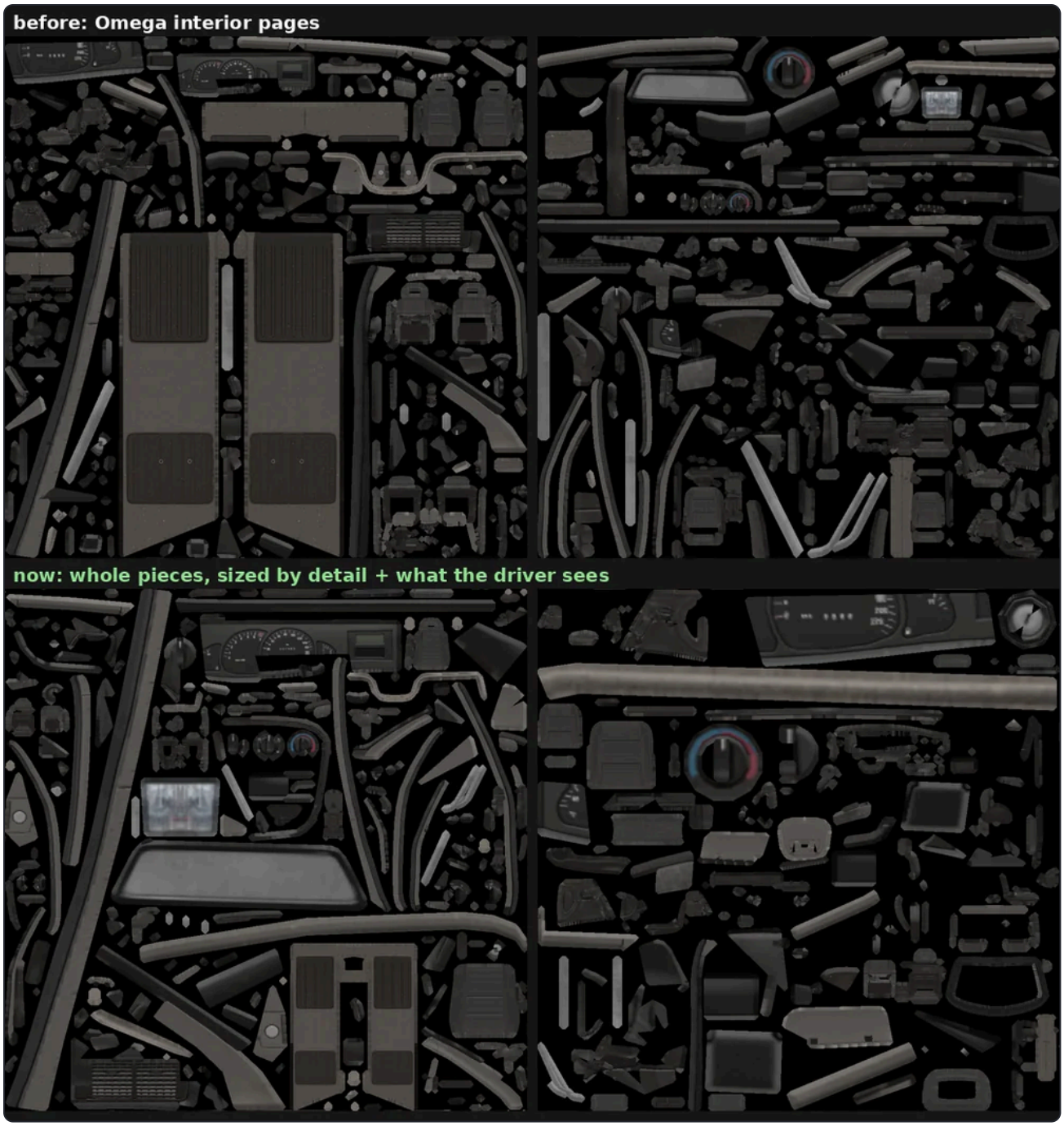


Fig. 25 The Omega's interior pages. Top: an older version cut every object into scraps by which way its faces point. Bottom: now whole pieces, with the gauges and the climate dial big and the floor mats small.



Fig. 26 A 3ds Max car whose outside came over as its left half: after Auto Setup, LFS mirrors it and the empty copies of the right doors and two wheels are filled.

EVERY BUTTON

Tool reference

Hover any button in Blender for its full tooltip. Every tool can be undone with `Ctrl` + `Z`.

BUTTON	WHAT IT DOES
LFS SRE panel	
Export to LFS	Exports the scene. Needs exactly one body.
Find Bad Triangles	Finds broken triangles that came with a file: zero-area ones and long thin slivers.
Rotate 180	Turns every marked part half a turn, for a car that faces backwards.
Show Mirrored Side	Shows the other side of the car the way LFS draws it (the mirror image of every face marked mirrored, and parts marked mirrored) in Solid view and while you model. Only for the eye: nothing is exported or changed. An import turns it on; press it to hide or show it.
Copy Bug Report	Copies versions, settings and the last messages, to paste where you ask for help.
1 Auto Setup	Sorts a model's meshes into body, interior, wheels and steering wheel, joins and marks them; finds textures that are not where the `.blend` says, stands up a car that came in on its end, turns a car that faces backwards round, gives parts without a material one by name and leaves out a floor, a studio panel or scenery. Shown for models from other formats.
2 Bake Textures for LFS	Makes the model's textures work in LFS: texture pages in `data\png`, the paint in a skin in `data\skins`, and an LFS map for every surface. Run it after Auto Setup, before export.
Undo Bake	Puts back the model's own materials after Bake Textures for LFS.
Models from other formats (closed section of the LFS SRE panel)	

BUTTON	WHAT IT DOES
Mark Selected Part(s)	Turns the selected meshes into new sub-objects. On an imported car use Join Selected Into… instead.
Prep Steering, Spin	Lays a steering wheel flat with its origin at the rim centre, then turns its spokes.
Purge Unused Maps	Deletes map entries no face uses and renumbers the rest.
Weld Panel Gaps	Closes small gaps between neighbouring panels of the active object.
Mark Double-Sided	Finds faces seen from both sides and makes the export write a reversed twin.
Fix Inside-Out Shells	Turns closed shells whose faces point inward.
Set Tile Size	Sets one tile size for every split map whose material name matches, so leather tiles line up.
Set Texture by Name…	Gives every material whose name contains a word the same LFS texture page.
Auto Split All Maps	Splits every sizeable tiling material by facing direction, except the body.
Split Map by Direction	Splits the active material’s map into top, bottom, front, back and sides, so textures don’t smear on curves.
Pro-Map Material	Maps the active material with small tiles aligned to each part, like hand-made interiors.
LFS Map per Material and LFS Texture per Material	
Auto Assign, Reassign All	Gives every material the LFS map with the most similar name; Reassign All also redoes the ones already set.
Clear All	Takes the map choice off every material.
Texture to Face Colours	Turns the active mesh’s texture into up to about 250 plain face colours.
Texture picker	Chooses the texture a material gets in LFS: a page from `data\png`, a plain colour, or automatic.

BUTTON	WHAT IT DOES
SRE Sub-Objects	
Type button, ×	Change a part's type (a dialog), or unmark it.
Select Added Faces	Selects the faces modelled since the import.
Join Selected Into...	Joins the selected new meshes into the part you pick, so they export with it.
Dish / Curve Part	Curves a part in or out. See Dish / Curve Part.
Smoothing groups	Show, set and preview groups. See Smoothing groups.
SRE Import and LFS Look	
Import SRE...	The same as **File › Import › LFS SRE (.sre)**.
Show LFS look, Editable model	Switch between the look and the model you edit.
Build / Rebuild LFS look	Builds the look again from the model as it is now.
Lighting	Sun turn, height, strength and colour, shadow fill, sky reflections, body colours.

TROUBLESHOOTING

When something is off

Textures are missing

Keep the LFS folder layout, with `data\3dob` beside `data\png`, or set **PNG Texture Folder** in the add-on's preferences. Then import again. SRE Import shows the texture folder in use.

Material Preview shows a plain model, not the look

The look may still be building (SRE Import says so), or it was switched off when you imported. Click **Build LFS look** in SRE Import. Also check *Show the LFS look* in the add-on's preferences.

Undo took the LFS look away

The look has its own step, *LFS look*, in **Edit › Undo History**. Undoing past it goes back to before the look was built, and SRE Import says *Undo removed the LFS look*. Redo with `Ctrl` `Shift` `Z`, or click **Build LFS look**.

A dished wheel shows a second rim behind it

Dish with Zone *Centre* moved the hub and the inner spokes back, out of the rim. Undo, and dish again with Zone *Ring*: the spokes bow in, the hub and the rim's edge stay.

The editor says "points have bad normals"

Usually triangles that stand exactly on the centre line of a mirrored part, often the wall an extrusion leaves along the centre. The mirror copy of such a triangle is the same triangle facing the other way, so their normals cancel. Delete them and export again. The export already warns about it, with the part and the number of points.

Every layer shows when I open the export in the modeller

The modeller does not save which layers are shown, whoever saved the file. The vehicle's configurations keep that, in its `.veh`: load the model into the vehicle and each configuration shows its own layers again.

The export has two cars in it

An export writes every imported part in the scene, and importing a second car adds its parts beside the first (the importer warns about it). Delete the old car's parts first, or import into a new scene.

The body is black or dark after a bake

The paint went into a skin, and colour 1 tints it. In the LFS look, set *Body colour* to white. In the vehicle editor, pick the skin and a white colour 1 in the *Colours* tab. A car whose paint has no texture of its own keeps a plain paint colour.

Bake packs all my textures into one picture

Tick **A texture page for each part** in the Bake dialog (on by default): the body, the wheels, the interior and the steering wheel each get pages of their own, named after them. Off, every part shares the same pages.

Bake cut my own textures up

Textures you made for LFS yourself: tick **Keep my texture layout** in the Bake dialog, bake again and export. Your pages go to LFS as they are; only glass and lamp lenses get a page of their own.

One side of the car is open in LFS

The model's outside is only one half of the car (a 3ds Max model that was mirrored in Max). Open the original model again and run **Auto Setup** with version 2.0.5 or newer: it finds the half and LFS mirrors it to the other side. Doors and wheels that came over as empty copies are filled too.

The editor shows a part off centre

Since 2.0.5 the body, the interior and the wheels export at position 0, 0, 0, so their centre line is the car's. Only the steering wheel keeps its own position, because LFS turns it around that point. Export again with the new version.

Blender closed during Bake

Fixed in 2.0.5: a bake that had to start again with the model's own textures could crash Blender. Update the add-on. If it still happens, click **Copy Bug Report** and send it with the model's name.

Bake says the texture folder is not set

Pick your LFS `data\png` folder in the box in the LFS SRE panel, or set **PNG Texture Folder** in the add-on's preferences, then bake again.

The model has no materials at all

A game rip or STL parts often come without any. **Auto Setup** gives each part one by its name – *auto glass* (see-through), *auto lens*, *auto tyre*, *auto rim*, *auto carbon*, *auto chrome*, *auto black plastic* and so on; body panels and names it cannot read become the paint. Its report says how many of each. Change any that are wrong in the Material tab, then run **Bake Textures for LFS**.

The model is pink in Material Preview

Its textures are not where the `.blend` says (a downloaded model often points at a `textures` folder that is not there). **Auto Setup** and **Bake Textures for LFS** look for them by name next to the `.blend`, in its sub-folders and the folder above, and say how many they found. Textures they still miss come out as a plain colour: use `File › External Data › Find Missing Files`, then bake again.

Blender did not tell me about an update

Blender checks a repository at startup only when its *Check for Updates on Startup* is ticked, and it is off when a repository is added. Tick it in the add-on's preferences, or in `Edit › Preferences › Get Extensions › Repositories` (top right): pick *byemk.github.io*. On the next start the status bar says *Extension Update Available*. Without it, open Get Extensions: the update shows there with an **Update** button.

The car faces backwards in LFS

Click **Rotate 180** and export again. (An older .blend may have *Global Y-flip* switched on: the LFS SRE panel then shows it in red; click it to switch it off.)

Export to LFS is grey

The scene needs exactly one body (type 0). The LFS SRE panel says *No Body!* or how many bodies it found. Part files without a body export from `File › Export`.

"Map table too long"

The vehicle editor crashes past about 236 maps. Run **Purge Unused Maps**, or run Bake Textures for LFS again (it stays under the limit). On an imported car, add parts with **Join Selected Into...**.

Part of the car is invisible from one side in LFS

LFS draws faces from one side. Use **Fix Inside-Out Shells** for closed shapes, or **Mark Double-Sided** for thin sheets.

A new part shades with odd hard edges

Click **Show** under Smoothing groups to see what the export will write. In Edit mode, **Join around** blends faces into their neighbours; **New group** gives a clean hard edge all round. **Shade like LFS** previews the result.

The LFS tab shows a red warning

Another copy of the LFS SRE add-ons is switched on, usually the older single-file exporter or importer. Turn it off in `Preferences › Add-ons`; two copies register the same buttons.

Blender is slow or looks frozen

Watch the cursor: a number on it means an import, export or bake is still working. On a slower PC switch off *Build it on import* and *Follow your edits* for the LFS look in the add-on's preferences.

Asking for help

Click **Copy Bug Report** (bottom of the LFS SRE panel, or the add-on's preferences) and paste it with your question: it says which versions you have and what Blender reported.

The look takes a while to appear

Blender compiles shaders before it shows the look for the first time. Give it a few seconds; switching back and forth is quick after that. Right after an import the look may still be building.

ABOUT

Credits and licence

LFS SRE Tools is made by **Mk** (Discord `@byeeeeeeeeeeeeeeeeeeeeeeeeeeeeee`), with code and this guide written together with Claude.

It is free software under the GNU General Public License, version 3 or later. You may use it, share it and change it. If you share a changed version, share its source under the same licence. The full text is in the `LICENSE` file in the zip.

This is a fan-made tool. It is not made by or connected to the Live for Speed developers.

Somewhere in the source code a cat is waiting. If you find it, it tells you what to type.
